



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Department of Aeronautical Engineering

2023-2024 EVEN SEM

(2020-24 Batch)

VALUE ADDED COURSE

“Fundamentals of Drone Technologies”

08.01.2024 - 12.01.2024


COORDINATOR


H.O.D


PRINCIPAL
PRINCIPAL
M.A.M. SCHOOL OF ENGINEERING
SIRIGANUR, TIRUCHIRAPPALLI-621 105.



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Department of Aeronautical Engineering Academic Year 2023-2024 EVEN Semester

Department Circular	Date: 21.12.2023
Issuing Authority: HOD	Classification: Periodical
Subject : Value added course -Reg	

We are planning to conduct the Value Added Course **Fundamentals of Drone Technologies** in association with Jet Aerospace Research Centre for final year Aeronautical students. Hence all the Final year students are informed to register their names to the respective class coordinators for the value added course is going to be conducted on 08.01.2024-12.01.2024 (5 Days) during 9.30 am to 4.30 pm.(Total No of Hours: 40 Hours)

Topics to be covered:

Fundamentals of Drone Technologies

Faculty In-Charge:

Mr.Joseph Arun Prasath.V.P/ AP/ Aero

Schedule:

Venue: AB 105 & 109, MAMSE.

Course Code & Name: FDT - Fundamentals of Drone Technologies	
S. No	Topics to be covered
1.	Introduction to Drones
2.	Drone Design & Flight Mechanics
3.	Drone Components & Systems
4.	Drone Flight Training & Simulation
5.	Autonomous Navigation & Control

Circular to:

- All Staff members of Aeronautical,
- To be read in IV Year class,
- Copy to the Notice Board.

HOD/AERO

Head of the Department
Aeronautical Engineering
M.A.M.School of Engineering
Siruganur, Trichy - 621 105.

	M.A.M. SCHOOL OF ENGINEERING Accredited by NAAC Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai Siruganur, Trichy -621 105. www.mamse.in	
---	--	--

Flyer / Brochure



M.A.M. SCHOOL OF ENGINEERING

(Accredited by NAAC)

(Approved by AICTE, New Delhi // Affiliated to Anna University, Chennai)

Siruganur, Trichy - 621105



Department of Aeronautical Engineering

Organizes
Value Added Course on

Fundamentals of Drone Technologies

In collaboration with
Jet Aerospace Research Centre



For Final Year
Aeronautical Students

Date : From 08.01.2024 to 12.01.2024 Time : From 09.30 AM to 04.30 PM

Mr.V.P.Joseph Arun
Coordinator

Dr.K.Chandrasekaran
Head of the Department

Dr.P.Ranjith Kumar
Principal

NBA - C9 - Co Curricular & Extra Curricular Activities

NAAC - C5 - Co Curricular Activities



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Department of Aeronautical Engineering
Academic Year 2023-2024 Even Semester
List of Participants for Value Added Course
FDT – Fundamentals of Drone Technologies

Date: 08.01.2024-12.01.2024

S. NO	REG NO	NAME OF THE STUDENT	DEPT	YEAR	SIGN
1	812120101001	ABISHEK K	AERO	III	A. Bishek
2	812120101002	ABISHEK S	AERO	III	S. Bishek
3	812120101003	ADHITHYAN R	AERO	III	R. Adhithyan
4	812120101004	AMSTRONG S	AERO	III	S. Armstrong
5	812120101005	BHAVANI M	AERO	III	M. Bhavani
6	812120101006	DURAIMANI E	AERO	III	E. Duraimani
7	812120101007	HARINI P	AERO	III	P. Harini
8	812120101008	HARISVALAN A	AERO	III	A. Harisvalan
9	812120101009	KARMEGAM P	AERO	III	P. Karmegam
10	812120101010	KATHIRAVA BALAN V	AERO	III	V. Kathirava Balan
11	812120101011	LIDEESH K	AERO	III	K. Lideesh
12	812120101012	MANI BHARATHI M	AERO	III	M. Mani Bharathi
13	812120101013	MANIKANDAN M	AERO	III	M. Manikandan
14	812120101014	MANOJ KRISHNAN M	AERO	III	M. Manoj Krishnan
15	812120101015	MUTHUKUMAR M	AERO	III	M. Muthukumar
16	812120101016	NITHISHKUMAR G	AERO	III	G. Nithish Kumar
17	812120101017	PANDIMEENAL N	AERO	III	N. Pandimeenal
18	812120101018	PRIYADHARSHINI K	AERO	III	K. Priyadharshini
19	812120101019	RAGUL K	AERO	III	K. Ragul
20	812120101020	SAKTHIVEL K	AERO	III	K. Sakthivel
21	812120101021	SANJAY K	AERO	III	K. Sanjay
22	812120101022	SANKARANARAYANAN	AERO	III	N. Sankaranarayanan
23	812120101023	SATHISH S	AERO	III	S. Sathish
24	812120101024	SERRY LEELIS M	AERO	III	M. Serry Leelis
25	812120101025	SILAMBARASAN S	AERO	III	S. Silambarasan
26	812120101026	SUJAY S	AERO	III	S. Sujay
27	812120101027	THENMOZHIN	AERO	III	N. Thenmozhi
28	812120101029	VIKRAM Y	AERO	III	Y. Vikram
29	812120101030	VIMAL RAJ V	AERO	III	V. Vimal Raj
30	812120101031	VISHNU J	AERO	III	J. Vishnu

HOD/AERO

Head of the Department
Aeronautical Engineering
M.A.M. School of Engineering
Siruganur, Trichy - 621 105.



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Department of Aeronautical Engineering
Academic Year 2023-2024 (Even SEMESTER)

Date: 08.01.2024

Content of the Program: Fundamentals Of Drone Technologies

This course provides a comprehensive introduction to drone technologies, covering the fundamental principles of drone design, operation, and applications. Students will gain an understanding of the various components that make up drones, their functions, and the regulatory environment governing their use.

Course Outline:

Day 1: Introduction to Drones

- History and Evolution of Drones
- Overview of Drone Technologies
- Types of Drones (Fixed-wing, Rotary-wing, Hybrid)

Day 2: Drone Design & Flight Mechanics

- Aerodynamics of Drones
- Lift, Thrust, Drag, and Weight
- Stability and Control

Day 3: Drone Components and Systems

- Airframes
- Propulsion Systems
- Power Sources (Batteries, Fuel Cells)
- Avionics and Sensors



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Day 4: Autonomous Navigation & Control:

- Artificial Intelligence and Machine Learning in Drones
- Advanced Navigation and Obstacle Avoidance
- Integration with Other Technologies (IoT, 5G)

Day 5: Drone Flight Training & Simulation

- Drone flight control using simulators.
- Safety protocols for drone operations.
- Manual and autonomous drone piloting.
- Flight maneuvers and missions in a simulated environment.

HOD/AERO

Head of the Department
Aeronautical Engineering
M.A.M.School of Engineering
Siruganur, Trichy - 621 105.



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Course Plan & Schedule

Course Code: FDT

Course Name: Fundamentals of Drone Technologies

Date	Session	Hours	Title
08.01.2024	F.N	4 hrs	History and Evolution of Drones
	A.N	4 hrs	Types of Drones (Fixed-wing, Rotary-wing, Hybrid)
09.01.2024	F.N	4 hrs	Aerodynamics of Drones
	A.N	4 hrs	Lift, Thrust, Drag, and Weight
10.01.2024	F.N	4 hrs	Components and Systems
	A.N	4 hrs	Drone Assembly and Maintenance
11.01.2024	F.N	4 hrs	Autonomous Navigation & Control
	A.N	4 hrs	Commercial Applications (Agriculture, Delivery, Inspection)
12.01.2024	F.N	4 hrs	Flight Simulation Training
	A.N	4 hrs	Project and Assessment
Total: 6 Days	10 Sessions	40 Hours	

HOD/AERO

Head of the Department
Aeronautical Engineering
M.A.M.School of Engineering
Siruganur, Trichy - 621 105.



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Snapshots of Sessions





M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

VAC – EVALUATION SHEET

Name of the Student: K.Ragul

Register Number: 812120101019

Sem / Year / Dept: VIII Sem / IV Year AERO

Mark Allotment			
S. No	Description	Allotted Marks	Obtained Marks
1	Drone Quiz	10	9
2	Group Discussion	10	9
3	Drone Component Identification	20	18
4	Drone Frame Assembly	20	19
5	Project Assignment	10	10
6	Flight Simulation	30	28
	TOTAL	100	93

Grade: *Excellent*


Trainer



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

FDT- Fundamentals of Drone Technologies VALUE ADDED COURSE - EVALUATION SHEET

S. No	Reg. No	Name	I	II	III	IV	V	VI	Total Mark	Grade
			10	10	20	20	10	30	100	
1	812120101001	ABISHEK K	9	9	18	20	9	27	92	Excellent
2	812120101002	ABISHEK S	7	9	15	15	7	22	75	Good
3	812120101003	ADHITHYAN R	8	8	16	16	8	25	81	Very Good
4	812120101004	AMSTRONG S	8	8	18	16	8	24	82	Very Good
5	812120101005	BHAVANI M	9	9	19	19	9	28	93	Excellent
6	812120101006	DURAIMANI E	9	8	18	16	10	27	88	Very Good
7	812120101007	HARINI P	7	7	15	15	7	23	74	Good
8	812120101008	HARISVALAN A	9	8	17	17	8	26	85	Very Good
9	812120101009	KARMEGAM P	9	10	18	20	9	29	95	Excellent
10	812120101010	KATHIRAVA BALAN	9	9	19	18	9	28	92	Excellent
11	812120101011	LIDEESH K	7	7	18	15	8	22	77	Good
12	812120101012	MANI BHARATHI M	9	9	19	19	9	29	94	Excellent
13	812120101013	MANIKANDAN M	8	8	16	16	9	24	81	Very Good
14	812120101014	MANOJ KRISHNAN	7	7	15	15	7	23	74	Good
15	812120101015	MUTHUKUMAR M	9	9	17	17	8	26	86	Very Good
16	812120101016	NITHISHKUMAR G	7	7	15	18	7	23	77	Good
17	812120101017	PANDIMEENAL N	9	9	18	18	9	27	90	Excellent
18	812120101018	PRIYADHARSHINI K	8	8	16	16	8	24	80	Very Good
19	812120101019	RAGUL K	9	9	18	19	10	28	93	Excellent
20	812120101020	SAKTHIVEL K	7	7	15	15	7	23	74	Good
21	812120101021	SANJAY K	8	8	16	16	9	24	81	Very Good



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

22	812120101022	SANKARANARAYANAN K	7	7	15	15	9	23	76	Good
23	812120101023	SATHISH S	8	8	17	17	8	26	84	Very Good
24	812120101024	SERRY LEELIS M	7	9	16	16	10	28	86	Very Good
25	812120101025	SILAMBARASAN S	8	8	16	16	9	27	84	Very Good
26	812120101026	SUJAY S	7	7	15	15	7	22	73	Good
27	812120101027	THENMOZHI N	8	8	16	16	8	25	81	Very Good
28	812120101029	VIKRAM Y	6	9	17	17	8	27	84	Very Good
29	812120101030	VIMAL RAJ V	9	9	18	18	10	27	91	Excellent
30	812120101031	VISHNU J	8	8	17	17	8	26	84	Very Good

Grade	Marks Range	Count
Poor	Below 60	00
Average	60 – 70	00
Good	70 – 80	08
Very Good	80 – 90	14
Excellent	Above 90	08

Trainer

HOD/AERO

Head of the Department
Aeronautical Engineering
M.A.M.School of Engineering
Siruganur, Trichy - 621 105.



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

TRAINER PROFILE

Name	:	Mrs,D.Arul Jyothi		
Designation	:	Administration Director		
Highest Qualification	:	M.E Aeronautical Engg		
Field of Interest	:	Unmanned Aerial Systems		
Company / Institution	:	Jet Aerospace Research Centre		
Contact Number	:	8883276477		
Email ID	:	info.jetaerospace@gmail.com		
Address	:	Near IIT Palakkad I-Hub, Opposite to ITI Ltd, Kanjikode Industrial Area, Palakkad - 678623, Kerala.		
Experience	:	8 years of Training Experience		
Profile	:	She has an experienced and certified drone training instructor with over 8 years of expertise in the field of unmanned aerial systems (UAS). Specializing in both commercial and recreational drone operations, She is dedicated to providing comprehensive and practical training programs to ensure pilots are proficient, safe, and compliant with all regulatory requirements. • Certified Instructor for DJI Enterprise Products • AUVSI Trusted Operator Program (TOP) Level 3 Certification • OSHA 10-Hour Construction Safety and Health • Designed and implemented comprehensive training modules for various skill levels. • Conducted over 500 hours of hands-on training sessions.		



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Course Outcome

1. Students gained a deep understanding of the fundamental principles of aerodynamics that govern drone flight. This includes the forces of lift, thrust, drag, and weight, and how they interact to affect a drone's stability and control.
2. Students were able to identify the various components that make up a drone, understand their functions, and integrate these components into a cohesive system.
3. Students learned to operate drones using remote control systems and understand the principles of autonomous flight, including the use of GPS and flight planning software.

HOD/AERO

**Head of the Department
Aeronautical Engineering
M.A.M.School of Engineering
Siruganur, Trichy - 621 105.**



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Feedback Form Report

Name of the Program:

VAC Program - Fundamentals of Drone Technologies

Date:

12.01.2024

- What is your opinion about the duration of this program?
☒ A. Short ☐ B. Adequate ☐ C. Long
- Overall, how useful was this program for you?
☐ A. Very Much ☒ B. To Some Extent ☐ C. Not Useful
- How would you rate the teaching Qualities?
☐ A. Very Good ☒ B. Good ☐ C. Average ☐ D. Poor
- How would you rate the materials presented?
☒ A. Very Good ☐ B. Good ☐ C. Average ☐ D. Poor
- How much of knowledge you learned today?
☒ A. A Lot ☐ B. Satisfactory ☐ C. None of it
- Did it fulfill your expectation?
☒ A. Yes ☐ B. Some Extent ☐ C. No
- Planning of this programme?
☒ A. Very Good ☐ B. Good ☐ C. Average ☐ D. Poor
- Any other comment (if any):



M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Feedback Form Report

Name of the Program:

VAC Program - Fundamentals of Drone Technologies

Date:

12.01.2024

- What is your opinion about the duration of this program?
☐ A. Short ☒ B. Adequate ☐ C. Long
- Overall, how useful was this program for you?
☒ A. Very Much ☐ B. To Some Extent ☐ C. Not Useful
- How would you rate the teaching Qualities?
☐ A. Very Good ☒ B. Good ☐ C. Average ☐ D. Poor
- How would you rate the materials presented?
☐ A. Very Good ☐ B. Good ☒ C. Average ☐ D. Poor
- How much of knowledge you learned today?
☒ A. A Lot ☐ B. Satisfactory ☐ C. None of it
- Did it fulfill your expectation?
☒ A. Yes ☐ B. Some Extent ☐ C. No
- Planning of this programme?
☐ A. Very Good ☒ B. Good ☐ C. Average ☐ D. Poor
- Any other comment (if any): NCL



M.A.M. SCHOOL OF ENGINEERING

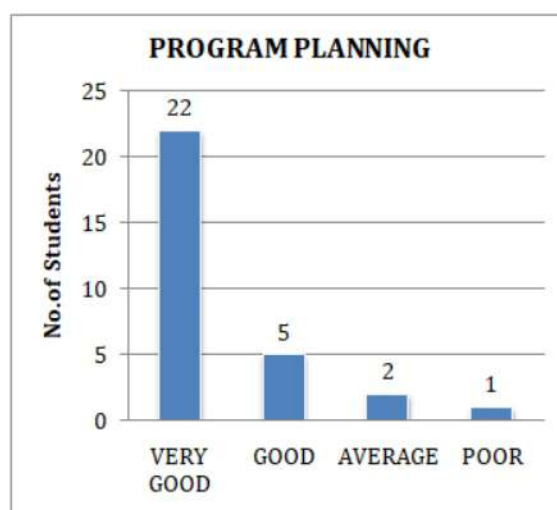
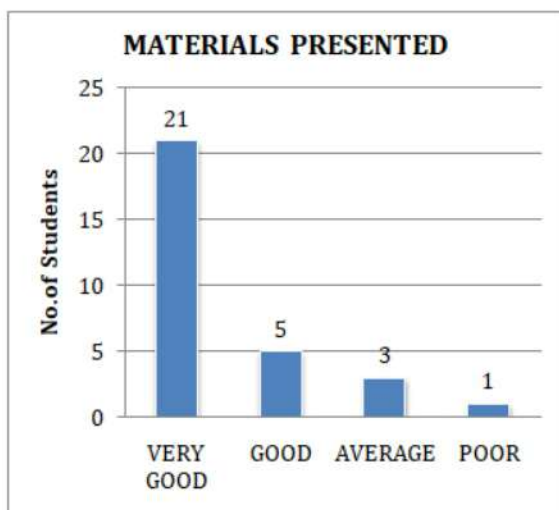
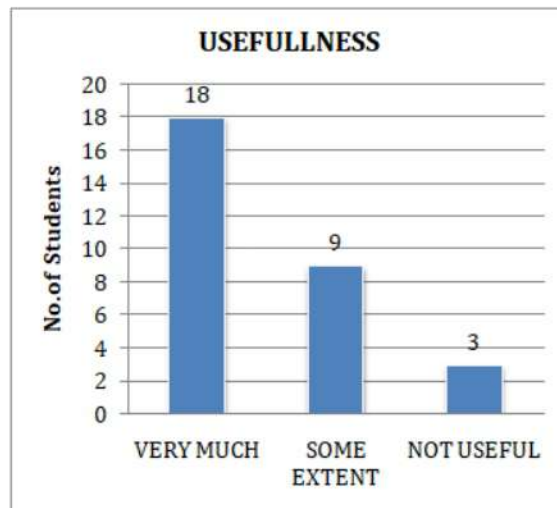
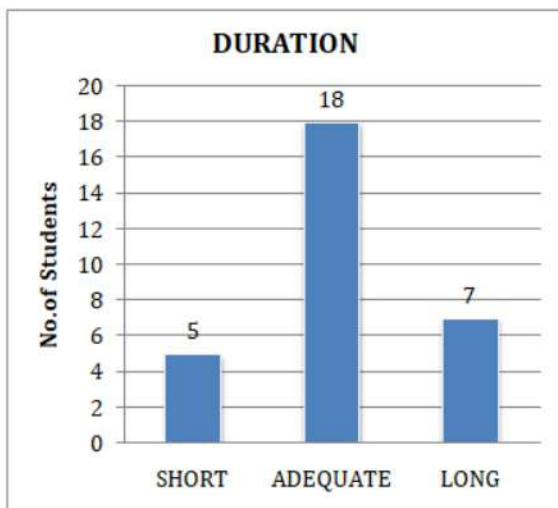
Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in

Feedback Summary





M.A.M. SCHOOL OF ENGINEERING

Accredited by NAAC

Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai

Siruganur, Trichy -621 105.

www.mamse.in



M.A.M. SCHOOL OF ENGINEERING

(Accredited by NAAC)

(Approved by AICTE, New Delhi // Affiliated to Anna University, Chennai)

Siruganur, Trichy - 621105



Certificate of Completion

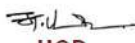
This is to certify that, K.Ragul of

IV Year Aeronautical Engineering branch has successfully completed the course "**Fundamentals of Drone Technologies**"

from **08.01.2024** to **12.01.2024** in association with Jet Aerospace

Score: 93 Grade: Excellent


Coordinator


HOD


Principal



M.A.M. SCHOOL OF ENGINEERING

(Accredited by NAAC)

(Approved by AICTE, New Delhi // Affiliated to Anna University, Chennai)

Siruganur, Trichy - 621105



Certificate of Completion

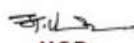
This is to certify that, S.Sujay of

IV Year Aeronautical Engineering branch has successfully completed the course "**Fundamentals of Drone Technologies**"

from **08.01.2024** to **12.01.2024** in association with Jet Aerospace

Score: 73 Grade: Good


Coordinator


HOD


Principal